

Document Recognition System

Modern document-reading systems combine optical character recognition with layout analysis. OCR identifies printed words, while layout models determine how information is arranged on the page.

Tables require special processing because their meaning depends on rows, columns and headers. The following table summarizes the main processing modules used in a typical intelligent reading pipeline.

Module	Function	Output
PaddleOCR	Recognizes printed text	Text
Table Detector	Finds table regions	Bounding box
Structure Model	Finds rows and columns	Table layout
RoBERTa	Answers text questions	Answer

Table 1. Main modules of the document-processing pipeline.

System Operation

After a page is captured, OCR extracts the visible words. If a table is detected, the table region is processed separately so that individual values remain associated with the correct headings.

This structured representation can later be translated, searched or converted into speech. Preserving the document structure therefore makes spoken output easier for a user to understand.

Experimental Evaluation

Image quality has a direct effect on document recognition. Good focus and lighting generally improve text recognition, while perspective distortion, page curvature and motion blur can reduce accuracy.

The table below presents example observations for several capture conditions. These values are provided as test content for evaluating OCR and table-structure recognition.

Condition	OCR	Table	Structure
Clear page	97.4%	Detected	Excellent
Perspective	93.1%	Detected	Good
Curved page	88.7%	Detected	Moderate
Motion blur	81.9%	Partial	Poor

Table 2. Example recognition results under different conditions.

Discussion

A table detector first identifies where the table is located. A structure-recognition model then identifies rows, columns and headers. OCR can finally associate recognized words with the correct cells.

This approach is useful for wearable document readers because the system can speak information row by row instead of reading unrelated words in an incorrect order.